

# Timestamp & Validation-Based Protocols DBMS

> **Definition:** Timestamp Ordering Protocols ensure conflict serializability without using locks by assigning a unique monotonic timestamp **TS(T<sub>i</sub>)** to each transaction upon start and enforcing operation ordering according to timestamp sequence.

## 1. Detailed Technical Explanation

### 1. System Timestamps

Each data item **X** maintains two timestamps:

### 2. Basic Timestamp Ordering Rules

#### Read Operation **Read(X)** by Transaction **T<sub>i</sub>**:

#### Write Operation **Write(X)** by Transaction **T<sub>i</sub>**:

## 2. Validation-Based (Optimistic) Protocol

Used when read operations dominate write operations (low conflict rate). Execution is split into 3 phases:

### Validation Test Condition:

For all **T<sub>k</sub>** committed before **T<sub>i</sub>**:

## 3. Core Concepts & Memory Keywords

- **Monotonic Timestamps:** Unique growing values (System clock or global counter).

#### 4. Must-Write Points for Exams

- Timestamp ordering protocols guarantee \*\*freedom from deadlocks\*\* because transactions are never made to wait indefinitely.

- Sta

---

#### 5. Quick Recall Flow

---

Comp